

Nihal Manjunath

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EDUCATION

BMS College Of Engineering, Bangalore
B.E. in Computer Science and Engineering

2022-2026
CGPA:7.3

TECHNICAL SKILLS

Languages: C++, TypeScript, JavaScript, Python, SQL
Frameworks: React, React Native, Node.js, Express.js, Tailwind CSS
AI/ML: RAG, pgvector, LLM APIs, LangChain, Function Calling, Prompt Engineering
Databases: PostgreSQL, MongoDB, MySQL, Redis, Firebase
Infrastructure: Docker, Kafka, Socket.IO, CI/CD, Vercel, Railway, Git/GitHub
Domains: Full-Stack Development, Distributed Systems, Microservices, Agentic AI, App Development

EXPERIENCE

Forward Deployed Engineer Intern | *Plivo* - [Link](#) Jan 2026 – Present

- Engineered AI-powered voice agents for enterprise clients using **Plivo Voice APIs**, **ElevenLabs** for **text-to-speech**, and **LiveKit** for **real-time audio streaming**, enabling automated customer interaction workflows at scale
- Built and deployed **Node.js scripts and automation tools** to streamline enterprise client onboarding and API integration, **reducing manual effort** across complex communication infrastructure.
- Diagnosed and resolved critical production incidents by querying **OpenSearch** and **Amazon Redshift** for **log analysis**, using **LogRocket** for **session replay** and **QuickSight** for **usage analytics** to identify root causes in API failures and call quality degradation.
- Served as primary technical liaison for enterprise clients, translating business requirements into production-grade API integrations and maintaining strong relationships across fintech and SaaS verticals.

PROJECTS

FDE Agent | *Python, RAG, LLM APIs, PostgreSQL/pgvector, Gmail/Calendar APIs, Slack, Redshift* [Demo](#) | [GitHub](#)

- Designed and shipped a production LLM email agent processing a live customer inbox across 6 workflow types (support, scheduling, pricing, debugging, follow-ups), with a structural human-approval gate. No auto-send path exists in the system, **1,300+ drafts** processed with zero unapproved outbound actions
- Built **anti-fabrication architecture** ("grounded-or-hold"): customer-facing claims trace only to retrieved documentation or the account's own warehouse records, enforced by code-level groundedness gates rather than prompts, after empirically demonstrating the model violating explicit prompt instructions
- Engineered exactly-once processing under unreliable delivery: idempotency anchors, atomic case claims, and reconciliation loops, validated live through real network flaps, dropped WebSocket events, and a full database outage with zero lost messages

FinFlow | *TypeScript, Node.js, PostgreSQL, Redis, Kafka, React* [Link](#) | [GitHub](#)

- Architected a production-grade fintech platform with **5 Node.js microservices** featuring a **PostgreSQL double-entry ledger with database-enforced immutability**, eliminating balance corruption bugs by construction
- Implemented distributed transaction safety using **Redis SETNX locks** with TTL ahead of SELECT FOR UPDATE, preventing race conditions across horizontally scaled service instances
- Built a real-time fraud detection engine with 4 independent rules - velocity tracking via Redis INCR, amount anomaly via PostgreSQL aggregates, large transaction thresholds, and round-trip detection via time-window queries

FlowDesk | *TypeScript, Node.js, PostgreSQL, Redis, Kafka, React* [Link](#) | [GitHub](#)

- Built a multi-tenant SaaS customer support platform with **6 Node.js microservices** behind an API gateway handling JWT auth, role-based access, and Redis sliding-window rate limiting, with tenant isolation enforced on every PostgreSQL query.
- Implemented **real-time chat and presence** using **Socket.IO** with Redis pub/sub for multi-instance broadcast, including live toast notifications for messages, assignments, and ticket events.
- Designed an event-driven pipeline over Kafka where services publish and return immediately while notification and analytics consumers process asynchronously, with no synchronous calls between services in the request path.

BMS College Canteen App (vk) | *React Native, Firebase, Expo, REST APIs* [Link](#) | [GitHub](#)

- Built and deployed a cross-platform food ordering app** serving **1,000+ students**, live on the Play Store, with real-time Firestore database management, authentication, and order tracking.
- Implemented **dynamic menu categories, cart management, and order tracking**, nested **Firestore subcollections** to manage food items and availability, ensuring seamless user interaction and stock synchronization.